

Debao Huang

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Education

Ph.D. , Electrical and Computer Engineering, The Ohio State University, USA	08/2026
M.S. , Electrical and Computer Engineering, The Ohio State University, USA	05/2023
B.S. , Telecommunication Engineering, Sun Yat-sen University, China	06/2019

Technical Skills

Programming Languages: C++, Python, MATLAB
Skills: Large-scale 3D Reconstruction (SfM, NeRF, 3DGS, 4DGS), Vision Foundation Models, Generative AI, Multimodal Learning
Frameworks and Libraries: PyTorch, Diffusers, Nerfstudio, PEFT, Docker, OpenCV, Ceres, COLMAP

Professional Experience

Research Scientist , Spectral Labs, San Francisco, CA	05/2026–Present
• Building foundation models for spatial intelligence that generate 3D geometry for engineering physical systems.	

Research Experience

Intrinsic Image Decomposition for Realism Enhancement of Digital Twins and 3D Assets – ONR	01/2025–04/2026
• Constructed a 33K+ image real-world dataset with pixel-aligned albedo, shading, normal, and metric depth using a physics-based inverse-rendering pipeline, enabling supervised intrinsic image decomposition.	
• Fine-tuned Stable Diffusion models using LoRA to close sim-to-real gap, improving PSNR by 7 dB over SOTA and enabling downstream applications such as relighting and material editing. (outcomes: P1: CVPR 2026).	
Live 3D Gaussian Splatting for Intelligence, Surveillance, and Reconnaissance – AFRL	10/2024–10/2025
• Developed a novel 4DGS pipeline to reconstruct dynamic urban scenes from monocular UAV video, achieving an improvement of 4 dB in PSNR over SOTA for rendering moving objects.	
• Integrated 4DGS with photogrammetry, video segmentation and tracking, monocular depth estimation, and physics-guided trajectory optimization into a unified reconstruction framework. (outcomes: P3).	
Enabling Seamless 3D Semantic Reconstruction from Heterogeneous Data at Scale – ONR	01/2021–02/2025
• Developed SfM algorithms and introduced novel geometric constraints in bundle adjustment for multi-camera systems, enhancing the accuracy of camera pose estimation by up to 86%. (outcomes: S1, H1, P4).	
• Developed a scalable Multi-Camera Tiling (MCT) NeRF framework for large-scale aerial datasets, reducing VRAM usage by 15.2% and significantly improving geometric completeness for fine structures. (outcomes: P5).	

Software & System Development

[S1] Software: MetricSfM (Lead Developer) C++, OpenMP, CUDA, Ceres, Eigen	2024
• Engineered an end-to-end Structure-from-Motion pipeline supporting heterogeneous data sources (aerial, UAV, GoPro, mobile, and underwater cameras) with both GUI and CLI interfaces. Deployed to the Office of Naval Research .	
[H1] Hardware: Multi-camera Mobile Mapping Systems	2022
• Developed a low-cost ground-mapping system using arbitrarily positioned GoPro cameras, incorporating geometric constraints for multi-camera self-calibration in MetricSfM to achieve robust 3D reconstruction.	

Selected Publications

[P1] Song, S. [†] , Huang, D. [†] , Deng, D., Xiong, H., Tang, Y., Zhao, Y., & Qin, R. (2026). “Olbedo: An Albedo and Shading Aerial Dataset for Large-Scale Outdoor Environments”. <i>The IEEE/CVF Conference on Computer Vision and Pattern Recognition</i> . (CVPR 2026).	
[P2] Huang, D. , Qin, R. (2026). “Uncertainty Quantification Framework for Aerial and UAV Photogrammetry through Error Propagation”. <i>ISPRS Journal of Photogrammetry and Remote Sensing</i> . (Top-tier journal, IF:12.2).	
[P3] Huang, D. [†] , Liu, H. [†] , Xu, N., & Qin, R. (2025). “Dynamic Urban Scene Modeling with 3D Gaussian Splatting from UAV Full Motion Videos”. <i>ISPRS Geospatial Week</i> .	
[P4] Huang, D. , Qin, R., & Elhashash, M. (2024). “Bundle Adjustment with Motion Constraints for Uncalibrated Multi-camera Systems at The Ground Level”. <i>ISPRS Journal of Photogrammetry and Remote Sensing</i> . (Top-tier journal, IF:12.2).	
[P5] Xu, N., Qin, R., Huang, D. , & Remondino, F. (2024). “Multi-tiling Neural Radiance Field (NeRF)—Geometric Assessment on Large-scale Aerial Datasets”. <i>The Photogrammetric Record</i> . (Cover article of 12/2024 issue).	

Honors & Media Coverage

- Recipient of **Lyman J. Ladner Memorial Scholarship**, American Society for Photogrammetry and Remote Sensing, 2026. Awarded to outstanding students (**Top 1%**) pursuing research and careers in photogrammetry.
- Research on 3D reconstruction using satellite data for disaster response, featured in **Ohio State News** [Link].